Ch 6

survival function of X : $S(t) = Pr(X > t) = 1 - F(t)$

$F(t) \Rightarrow Pr(X \le t)$

hazard func. / mortality func. / failure func. :

$$\lambda(t) = \frac{f(t)}{S(t)}$$

Remark:

$$\underline{Pr(t < X < t+dt \mid X > t)} = \frac{F(t+dt) - F(t)}{S(t)} \approx \frac{f(t)\,dt}{S(t)} = \underline{\lambda(t)\,dt}$$

$\lambda(t)$ unique ~ $F(t)$

$$\underline{f(t) = F'(t) = S'(t)}$$

$$\underline{\lambda(t) = \frac{f(t)}{S(t)} = \frac{S'(t)}{S(t)} = -\frac{d}{dt}[\ln S(t)] = -\frac{d}{dt}[\ln(1-F(t))]}$$

$$\Rightarrow \int_0^x \lambda(t)\,dt = -\ln[1 - F(t)]$$

$(\int f' = f \;;\; g = f',\ \int g = f)$

E.g.: $X \sim Exp(\lambda) \Rightarrow f(t) = \lambda e^{-\lambda t} \Rightarrow F(t) = 1 - e^{-\lambda t}$

$$\Rightarrow S(t) = 1 - (1 - e^{-\lambda t}) \Rightarrow \lambda(t) = \frac{f(t)}{S(t)} = \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} = \lambda$$

E.g. 2: Linear hazard rate func.: $\underline{\lambda(t) = a + bt}$

(线性)

(对于这个 $\lambda(t)$:) $-\ln[1 - F(t)] = ax + \frac{b}{2}t^2$

$$\Rightarrow 1 - F(t) = e^{-(ax + \frac{b}{2}t^2)}$$

$\Rightarrow F(t) = \sim$, $f(t) = \lambda(t)[1 - F(t)]$

indicator function of set A ($A \subset \Omega$): $I_A: \Omega \to \{0, 1\}$,

$$I_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$

Properties:

$I_{A \cap B} = \min(I_A, I_B) = I_A \times I_B$

$I_{A \cup B} = \max(I_A, I_B) = I_A + I_B - I_A \times I_B$

$I_{A^c} = 1 - I_A$

$E(I_A) = Pr(A)$ $\left(\int_{-\infty}^{\infty} I_A f(x)\,dx = \int_A f(x)\,dx \right.$

Remark:

$I_A \sim Ber(Pr(A))$

$$f(x) = \begin{cases} \sim, & \text{for } x \in A \\ 0, & \text{otherwise} \end{cases} \iff f(x) = \sim \cdot I_{\{x \in A\}}$$

1 a func

$$X(w) = \begin{cases} c_1, & \text{if } w \in A \\ c_2, & \text{if } w \in B \\ c_3, & \text{if } w \in C \end{cases} \iff X(w) = c_1 \cdot I_A + c_2 \cdot I_B + c_3 \cdot I_C$$

$$= c_1^{I_A} \times c_2^{I_B} \times c_3^{I_C}$$

1 a r.v.

E.g.:

Markov's inequality: $(\forall \omega, X(\omega) \geq 0$ and $\mu < \infty$, $\mu = E(X))$

$\forall c,\ Pr(X \geq c) \leq \frac{E(X)}{c}$

Proof 1:

Proof 2:

Chebyshev's Inequality: $(\mu < \infty, \sigma^2 < \infty)$

$\forall k > 0,\ Pr(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$

Proof: $(E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] = \frac{1}{\sigma^2} \underbrace{E[(X-\mu)^2]}_{=\sigma^2} = 1$

$= E(X^2) - [E(X)]^2$

E.g.:

Dist. of Mixed Type

Quantiles (分位数)

X is cont. & X is one-to-one & the inverse dist. func. F^{-1} exists

$\Rightarrow$ the p-quantile of X for $0<p<1$ (the 100p percentile of X): $F^{-1}(p)$

(100×p, 需计算)

X is discrete

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|, \quad \forall y \in g(S)$$

($Y = g(X)$, X dist. on space S)

E.g.



Ch 7

joint pmf of X, Y : $p(x,y) = Pr(X=x, Y=y)$, $(x \in X(\Omega), y \in Y(\Omega))$

(X, Y: discrete; ",": (and))

Conditions:

$0 \le p(x,y) \le 1$, $\forall x \in X(\Omega)$ and $y \in Y(\Omega)$

$\sum_{x \in \cdots} \sum_{y \in \cdots} p(x,y) = 1$

$A \subset X(\Omega) \times Y(\Omega)$, $Pr((X,Y) \in A) = \sum_{(x,y) \in A} p(x,y)$

marginal pmfs of X : $p_X(x) = Pr(X=x) = \sum_{y \in Y(\Omega)} p(x,y)$

of Y : $p_Y(y) = Pr(Y=y) = \sim$

Remark:

joint pmf $\xrightarrow{\text{unique}}$ marginal pmf

$\not\leftarrow$

X and Y are jointly continuous (set)

$\Leftrightarrow \forall C \subseteq \mathbb{R}^2$ (measurable), $\exists$ (function) $f(x,y)$ s.t. $Pr\{(X,Y) \in C\} = \iint_{(x,y) \in C} f(x,y)\,dx\,dy$

→ joint pdf

Properties:

(见下页)

(接上页)

(Properties (of joint pdf)):

(joint dist. func.:

$F(x,y) = Pr(X \le x, Y \le y)$

$= \int_{-\infty}^{y} \int_{-\infty}^{x} f(x,y)\, dx\, dy$)

(7, 2)

((2 events) A and B are independent: $\Leftrightarrow Pr(A \cap B) = Pr(A) Pr(B)$

$\Leftrightarrow Pr(A|B) = Pr(A)$

(2 r. v.) X and Y are independent

$\Leftrightarrow$ { X & Y are discrete: ($\exists$ 必为 $\exists$, $\neq$ $\forall$ x and y, $\rightarrow p(x,y) = p_X(x) p_Y(y)$

X & Y are continuous; $\rightarrow f(x,y) = f_X(x) f_Y(y)$

$\Leftrightarrow$ { supports of X and Y do not depend on each other

and

$\exists g(x), h(y)$, s.t. $f(x,y) = g(x) h(y)$

Remark:

generalize joint ~ & marginal ~

Joint pmf of ~: $p(x_1, \cdots, x_n) = Pr(X_1 = x_1, \cdots, X_n = x_n)$

Joint pdf of ~: $f(x_1, \cdots, x_n)$

Marginal pmf of ~: $p_{X_1}(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, \cdots, x_n)$, $x_1 \in X_1(\Omega)$

Marginal pdf of ~: $f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, \cdots, x_n) f(x_1, \cdots, x_n)\, dx_1 \cdots dx_n$, $-\infty < x_i < \infty$

$X_1, \cdots, X_n$ are mutually indep.

$\Leftrightarrow$ discrete: $p(x_1, \cdots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$, $\forall x_1, \cdots, x_n$

continuous: $f(x_1, \cdots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$ $\forall$

$\Rightarrow \forall A_1, \cdots, A_n \subset (-\infty, \infty)$,

$Pr(X_1 \in A_1, \cdots, X_n \in A_n) = Pr(X_1 \in A_1) \cdots Pr(X_n \in A_n)$

$\Leftrightarrow$

$F(x_1, \cdots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n)$

Expectation

Discrete: $E[u(X_1, \cdots, X_n)] = \sum_{x_1} \cdots \sum_{x_n} u(x_1, \cdots, x_n) \underbrace{p(x_1, \cdots, x_n)}_{pmf}$

$\Downarrow$

Continuous: $= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, \cdots, x_n) \underbrace{f(x_1, \cdots, x_n)}_{pdf} dx_1 \cdots dx_n$

Properties:

$E[u_1(X) + u_2(Y)] = \sim + \sim$

X & Y are indep $\Rightarrow E[u_1(X) u_2(Y)] = \sim \cdot \sim$

$\Rightarrow M_{X+Y}(t) = \sim \cdot \sim$

$M_{aX+bY+c}(t) = M_X(at) \cdot M_Y(bt) \cdot e^{ct}$

(分布)

(7.4)

Population Covariance

$$\sigma_{XY} = Cov(X, Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

$$= \underline{E(XY)} - \mu_X\mu_Y$$

$$= \sum_{x=i_1}^{i_2}\sum_{y=j_1}^{j_2} xy \cdot p(x,y)$$

$$\text{or } \int_{\sim}^{\sim}\int_{\sim}^{\sim} xy \cdot f(x,y)\,dx\,dy$$

Properties

$\sigma_{XY} > 0 \Rightarrow X, Y$ 同升

$\sigma_{XY} < 0 \Rightarrow X, Y$ 异升

$Cov(aX+b, cY+d) = ac\,Cov(X,Y)$

$\forall$ constant C, $\forall$ r.v. X, $Cov(X, C) = 0$

$Cov(X, X) = Var(X)$

$$Cov(\underbrace{a_1X_1 + \cdots + a_nX_n}_{(n\text{个r.v.})}, \underbrace{b_1Y_1 + \cdots + b_mY_m}) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_ib_j\,Cov(X_i, Y_j)$$

$$Var(X_1 + \cdots + X_n) = Var(X_1) + \cdots + Var(X_n) + \underbrace{2\sum_{i<j} Cov(X_i, X_j)}$$

$$\sum_{i\neq j} Cov(X_i, X_j) =$$

E.g. X 2

population correlation coefficient between X and Y:

$$\rho = \rho_{X,Y} = Corr(X,Y) = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

$\sigma_X = \sqrt{Var(X)}$

$\sigma_Y = \sqrt{Var(Y)}$

Cauchy-Schwarz Inequality:

X, Y: r.v. with finite 2nd moments

$$[E(XY)]^2 \le E(X^2)E(Y)^2$$

$\iff Pr(X=0)=1$ or $Pr(Y=aX)=1$ ($\exists$)

(Proof: ② Consider $\forall t \in \mathbb{R}$, consider

$$0 \le E[(tX+Y)^2] = t^2E(X^2) + 2tE(XY) + E(Y^2)$$

)

Properties:

$-1 \le \rho \le 1$

$Corr(aX+b, cY+d) = \sim = sgn(ac)\,Corr(X,Y)$

X & Y are indep. $\Rightarrow Var(X+Y) = Var(X) + Var(Y)$

$\Rightarrow Cov(X,Y)=0 \Rightarrow \rho = 0$ $\quad Var(X-Y)$

$\Rightarrow$ X, Y are ~~not~~ constants

Remarks:

$\rho = 0 \not\Longrightarrow X, Y$: indep.

$\rho = 0 \Longleftarrow X, Y$: indep.

X and Y are uncorrelated

Correlation coeff. measures the strength of the linear relationship only

Ch 8

(discrete)

Conditional pmf of Y given that $X=x$:

$(\forall x, p_X(x) \geq 0)$

$$p_{Y|X}(y|x) = Pr(Y=y \mid X=x) = \frac{Pr(Y=y, X=x)}{Pr(X=x)}$$

$$= \frac{p_{X,Y}(x,y)}{p_X(x)} \longrightarrow \text{marginal}$$

$\longrightarrow$ joint

Y is indep. of $X \iff p_{Y|X}(y|x) = p(y)$

$$= \frac{p(y)p(x)}{p(x)}$$

(continuous)

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

Properties:

$$F_{Y|X}(y|x) = Pr(Y \leq y \mid X=x) = \begin{cases} \text{(discrete)} \sum_{i \leq y} p_{Y|X}(i|x) \\ \text{(continuous)} \int_{-\infty}^{y} f_{Y|X}(t|x)\,dt \end{cases}$$

$$E[g(Y) \mid X=x] = \begin{cases} \text{(d.)} \sum_i g(i)\, p_{Y|X}(i|x) \\ \text{(c.)} \int_{-\infty}^{\infty} g(t)\, p_{Y|X}(t|x)\,dt \end{cases}$$

$$Var[(Y|X=x)] = E[(Y - (E(Y|X=x))^2) \mid X=x]$$

$$= E(Y^2|X=x) - [E(Y|X=x)]^2$$

Law of Total Expectation

$$E[u(X)] = E[E(u(X)|Y)]$$

Adam's Law

Law of Total Variance

$$Var(X) = E[Var(X|Y)] + Var[E(X|Y)]$$

Moment & MGF

Ch 4

(Discrete)

the r-th moment of X (about b): $E[(X-b)^r]$

the r-th central moment of X: $E[(X-\mu)^r]$

mgf of X: $M_X(t) = \sum_{x \in X(\Omega)} e^{tx} p(x) = E(e^{tX})$

(Ω: space, p: pmf)

Properties

$M_X^{(r)}(0) \left(= \frac{d^r}{dx^r} M_X(t)\Big|_{t=0}\right) = E(X^r)$

$M_{aX+b}(t) = E(e^{(aX+b)t}) = e^{bt} M_X(at)$

$M_X(t) = M_Y(t) \Rightarrow X$ & Y have same distribution

Cgf of X: $R_X(t) = \ln(M_X(t))$

Properties:

$R_X^{(1)}(0) = E(X) = \mu$

$R_X^{(2)}(0) = Var(X) = \sigma^2$

$\frac{R_X^{(3)}(0)}{\sigma^3} = Skew(X)$

$R_X^{(4)}(0) \sim Kurt(X)$

$R^{(3)}(0) = E[(X-\mu)^3]$

pgf of X: $G_X(z) = E(z^X)$

$(= E(e^{\ln(z) \cdot X})) = M_X(\ln z)$

Properties:

Ch 5

(Continuous)

(Expectation: $E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx$

Properties

(mgf:)

$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$

$M_X^{(r)}(0) = E(X^r)$

(cgf:)

$R_X(t) = \ln(M_X(t))$

$R_X^{(1)}(0) = \mu$

$R_X^{(2)}(0) = \sigma^2$

Tut

T10:

Q3: (a) valid pdf $\Leftarrow$ $\begin{cases} \in [0,1] \\ \iint_R f_{X,Y}(x,y)\,dx\,dy = 1 \end{cases}$

(b) X, Y indep $\Leftrightarrow$ $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$

$f_{X,Y}$: joint; f_X: marginal; f_Y: marginal

Q1: $Pr(c < X \leq x) = Pr(X \leq x) - Pr(X < c)$